

Comparative Analysis of Classification Algorithms

Problem Statement

A diagnostic dataset contains measurements computed from digitized images of breast tissue samples. The objective is to develop a machine learning system that classifies a sample into one of two classes:

- **Benign**
- **Malignant**

Using the **Breast Cancer Wisconsin (Diagnostic) dataset from Kaggle**, develop and compare different machine learning classification algorithms.

Tasks to be performed

A. Data Understanding and Preprocessing

1. Download the dataset from Kaggle and load it using Pandas:
<https://www.kaggle.com/datasets/uciml/breast-cancer-wisconsin-data>
2. Identify:
 - Number of samples
 - Number of features
 - Target variable
 - Data types
 - Missing values
 - Class distribution
3. Perform appropriate preprocessing:
 - Remove irrelevant attributes such as ID if present.
 - Encode the target variable.
 - Check and handle missing values.
4. Analyze the distribution of the two classes.
5. Divide the dataset into: 80% Training+20% Testing
6. Use a fixed **random_state** and **stratified splitting**.
7. Apply feature scaling where appropriate.

B. Exploratory Data Analysis

Perform suitable exploratory analysis and answer:

1. Which features appear to have strong relationships with the target?
2. Are there highly correlated features?
3. Are there features with very different scales?
4. Are the two classes visually separable using selected features?

Generate the following plots:

- Class distribution plot, and comment on it; what do you observe from the plot?
- Correlation heatmap, and comment on it; what do you observe from the plot?
- Suitable feature distribution plots, and comment on it; what do you observe from the plot?
- Scatter plot using two important features, and comment on it; what do you observe from the plot?

C. Implement Classification Algorithm- Logistic Regression

1. Implement Logistic Regression and investigate the effect of the regularization parameter C .
2. Experiment with at least three values, for example: 0.01, 0.1, 1, 10
3. Evaluation of Classification Performance, calculate **Accuracy, Precision, Recall, F1-score, Confusion Matrix, ROC-AUC, Classification report, ROC Curve**
4. **Precision:** What does the precision value indicate about the model's ability to correctly identify the positive class? Which value of C gives the highest precision?
5. **Recall:** What does the recall value indicate about the model's ability to detect actual positive cases? Which value of C gives the highest recall?
6. **F1-score:** Which value of C provides the best balance between precision and recall? Why?
7. **Confusion Matrix:** Examine the confusion matrix for each value of C and identify the number of:
 - True Positives (TP)
 - True Negatives (TN)
 - False Positives (FP)
 - False Negatives (FN)

Which value of C produces the fewest misclassifications?

8. **ROC Curve:** Plot the ROC curves for all values of C on the same graph. Compare the curves and their **ROC-AUC** values. Which model provides the best class-separation capability?

9. **Effect of Regularization:** How does changing **C** affect precision, recall, F1-score, and ROC-AUC? Explain the observed behavior.
10. **Precision vs Recall:** Considering that this dataset represents a **medical diagnosis problem**, determine whether **precision or recall should be given greater importance**. Justify your answer in terms of **False Positives and False Negatives**.
11. **Final Model Selection:** Based on the confusion matrix, precision, recall, F1-score, and ROC-AUC, **which value of C would you select for deployment and why?** Do not select the model based only on accuracy.
12. Prepare a table of the following form:

C	Accuracy	Precision	Recall	F1-score	ROC-AUC	FP	FN
0.01							
0.1							
1							
10							

D. Implement Naive Bayesian Classification

(i) Interpret the obtained values and answer:

1. Is the model producing reliable positive predictions?
2. How effectively does it identify actual positive cases?
3. Does the F1-score indicate a good balance between precision and recall?
4. Which class is being classified most accurately?
5. Which class is most difficult for the model to classify?

(ii) Generate the **ROC curve** using the predicted probabilities.

(iii) Prepare a short analysis answering:

1. How well does Gaussian Naive Bayes perform on the dataset?
2. Which class has the highest precision?
3. Which class has the highest recall?
4. What does the F1-score tell you about the model?
5. What does the confusion matrix reveal that accuracy alone does not?
6. What does the ROC curve and ROC-AUC indicate?
7. Is the **Naive Bayes independence assumption** reasonable for this dataset?
8. What are the major advantages and limitations of using Naive Bayes for this problem?

E. Implement Decision Tree Classifier

(i) Perform hyper parameter tuning with criterion Entropy:

- max_depth
- min_samples_split
- min_samples_leaf
- criterion

(ii) What is the optimum set of hyper parameters

(iii) With optimum hyper parameters chosen in above step compare Gini and Entropy.

(iv) Whether increasing tree depth improves or reduces generalization.

F. Implement Random Forest Classifier

(i) Perform hyper parameter tuning with:

- N_estimators = 50, 100, 200
- Max_depth = 3, 5, 10
- max_features
- min_samples_split

(ii) What is the optimum set of hyper parameters

(iii) Using the **optimum hyperparameters obtained in Step (ii)**, train two Random Forest classifiers by changing only the splitting criterion:

- criterion = 'gini'
- criterion = 'entropy'

(iv) Whether increasing tree depth improves or reduces generalization.

(v) Does changing the splitting criterion significantly affect the classification performance?
Which criterion would you select and why?

(vi) How does increasing max_depth affect training accuracy?

(vii) How does it affect testing accuracy?

(viii) At what depth does the model begin to show signs of **overfitting**?

(ix) Does increasing tree depth always improve generalization? Explain using your experimental results.

Kindly note: This assignment need to be uploaded by 16th August 4 pm